

BLAGOVESHENKA, Kazakhstan

WMO: 287660

Lat: 54.3589N

Lon: 66.9801E

Elev: 495

StdP: 14.43

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.6	-22.6	-34.7	0.8	-28.4	-28.7	1.1	-22.3	27.8	16.4	26.9	17.6	2.4	230	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	87.8	65.9	84.2	65.1	81.0	63.9	69.2	81.3	67.6	79.1	66.1	77.0	7.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	94.8	73.6	63.7	89.8	72.2	62.1	84.8	70.8	33.6	81.3	32.3	78.9	31.1	77.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	26.7	22.1	18.5	DB	-34.2	92.7	6.4	4.2	-38.8	95.8	-42.5	98.2	-46.1	100.6	-50.8	103.6
(5)				WB	-34.5	72.3	6.2	1.6	-38.9	73.4	-42.6	74.3	-46.1	75.2	-50.6	76.3

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	0.7	3.8	19.3	39.8	55.3	64.3	66.4	63.4	52.9	39.3	21.4	7.0	(6)
(7)		DBStd	25.91	13.13	13.08	10.38	10.88	8.83	7.65	6.32	7.36	8.28	8.58	13.02	13.57	(7)
(8)		HDD50	6767	1529	1295	952	336	50	4	0	2	60	346	858	1335	(8)
(9)		HDD65	10762	1994	1715	1417	757	317	103	61	121	371	798	1308	1800	(9)
(10)		CDD50	1762	0	0	0	29	214	434	507	416	148	14	0	0	(10)
(11)		CDD65	282	0	0	0	0	15	83	103	71	10	0	0	0	(11)
(12)		CDH74	3045	0	0	0	8	294	935	962	706	140	0	0	0	(12)
(13)		CDH80	902	0	0	0	1	64	308	282	211	36	0	0	0	(13)
(14)	Wind	WSAvg	6.7	6.7	7.3	7.6	7.4	7.4	6.1	5.2	5.2	6.0	6.9	7.0	7.2	(14)
(15)	Precipitation	PrecAvg	14.3	0.7	0.5	0.7	0.9	1.1	1.8	2.5	1.9	1.3	1.1	1.0	0.8	(15)
(16)		PrecMax	20.6	1.3	1.3	1.6	2.6	2.6	5.9	5.0	5.3	2.6	2.4	2.0	1.9	(16)
(17)		PrecMin	8.0	0.0	0.0	0.0	0.1	0.1	0.3	0.6	0.1	0.2	0.1	0.1	0.2	(17)
(18)		PrecStd	3.0	0.4	0.3	0.4	0.6	0.6	1.2	1.3	1.2	0.6	0.6	0.5	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	34.2	46.6	74.3	86.2	91.3	91.6	90.4	84.5	68.6	47.8	34.1	(19)
MCWB			30.6	31.9	40.7	55.5	62.0	66.2	67.0	66.0	60.6	52.1	43.1	31.9	(20)	
2%		DB	28.3	30.7	37.8	67.0	80.9	87.6	86.4	85.9	77.9	62.5	42.5	31.7	(21)	
		MCWB	26.9	29.3	34.7	52.6	59.5	65.6	67.6	65.5	58.1	50.1	40.0	30.3	(22)	
5%		DB	25.1	27.4	35.5	62.1	77.0	83.4	82.8	81.6	71.9	57.3	39.4	28.5	(23)	
		MCWB	23.8	26.0	33.3	50.5	58.2	65.3	65.8	64.9	57.0	47.4	37.0	27.2	(24)	
10%		DB	20.9	23.4	33.6	56.9	72.4	79.3	79.4	77.6	66.8	52.6	36.1	25.0	(25)	
		MCWB	19.9	22.1	31.7	47.8	56.6	63.2	65.2	63.5	55.0	45.6	33.7	23.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.7	32.2	40.6	57.4	64.7	70.9	71.1	70.0	64.1	55.0	44.1	32.2	(27)
MCD B			32.0	33.9	46.4	69.6	78.6	84.4	82.8	82.1	79.1	66.3	46.4	33.4	(28)	
2%		WB	27.0	29.3	35.1	54.0	62.1	68.5	69.4	68.2	60.3	51.4	40.4	30.1	(29)	
		MCD B	28.2	30.7	37.1	63.8	75.9	81.3	81.2	80.1	72.2	59.6	42.5	31.9	(30)	
5%		WB	23.9	26.1	33.5	51.3	59.8	66.3	68.0	66.5	58.2	48.3	37.2	27.1	(31)	
		MCD B	25.0	27.4	35.3	61.1	73.3	79.8	79.1	77.2	69.2	55.7	39.2	28.4	(32)	
10%		WB	19.9	22.2	31.7	48.1	57.6	64.5	66.5	64.7	55.9	45.8	34.0	23.6	(33)	
		MCD B	20.9	23.5	33.6	56.9	70.3	76.4	76.5	74.6	65.5	52.1	35.6	24.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.0	15.5	15.4	17.1	21.6	20.3	18.4	19.2	18.9	14.7	11.8	13.5	(35)
MCD BR			13.4	13.0	12.8	25.7	28.9	27.0	24.5	26.2	27.1	24.7	12.1	12.1	(36)	
MCWBR			13.2	12.4	10.8	14.1	12.7	11.4	9.2	10.6	12.3	14.4	9.9	11.7	(37)	
5% WB		MCD BR	13.5	13.0	12.0	23.1	25.4	22.7	20.4	22.0	23.8	21.9	11.3	12.4	(38)	
		MCWBR	13.3	12.4	10.4	14.0	13.1	10.5	9.1	10.7	12.1	13.7	10.0	12.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.320	0.373	0.408	0.407	0.406	0.424	0.432	0.380	0.356	0.320	0.286	(40)	
taud		2.057	2.017	1.961	2.131	2.226	2.306	2.247	2.196	2.298	2.246	2.136	2.035	(41)		
Ebn at Noon		199	233	246	257	266	267	259	248	247	221	184	167	(42)		
Edh at Noon		22	33	44	43	41	39	40	40	32	26	21	18	(43)		
(44)	All-Sky Solar Radiation	RadAvg	333	681	1172	1466	1790	1927	1793	1463	1033	529	264	214	(44)	
(45)		RadStd	23	30	85	157	147	163	152	144	72	83	34	29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page